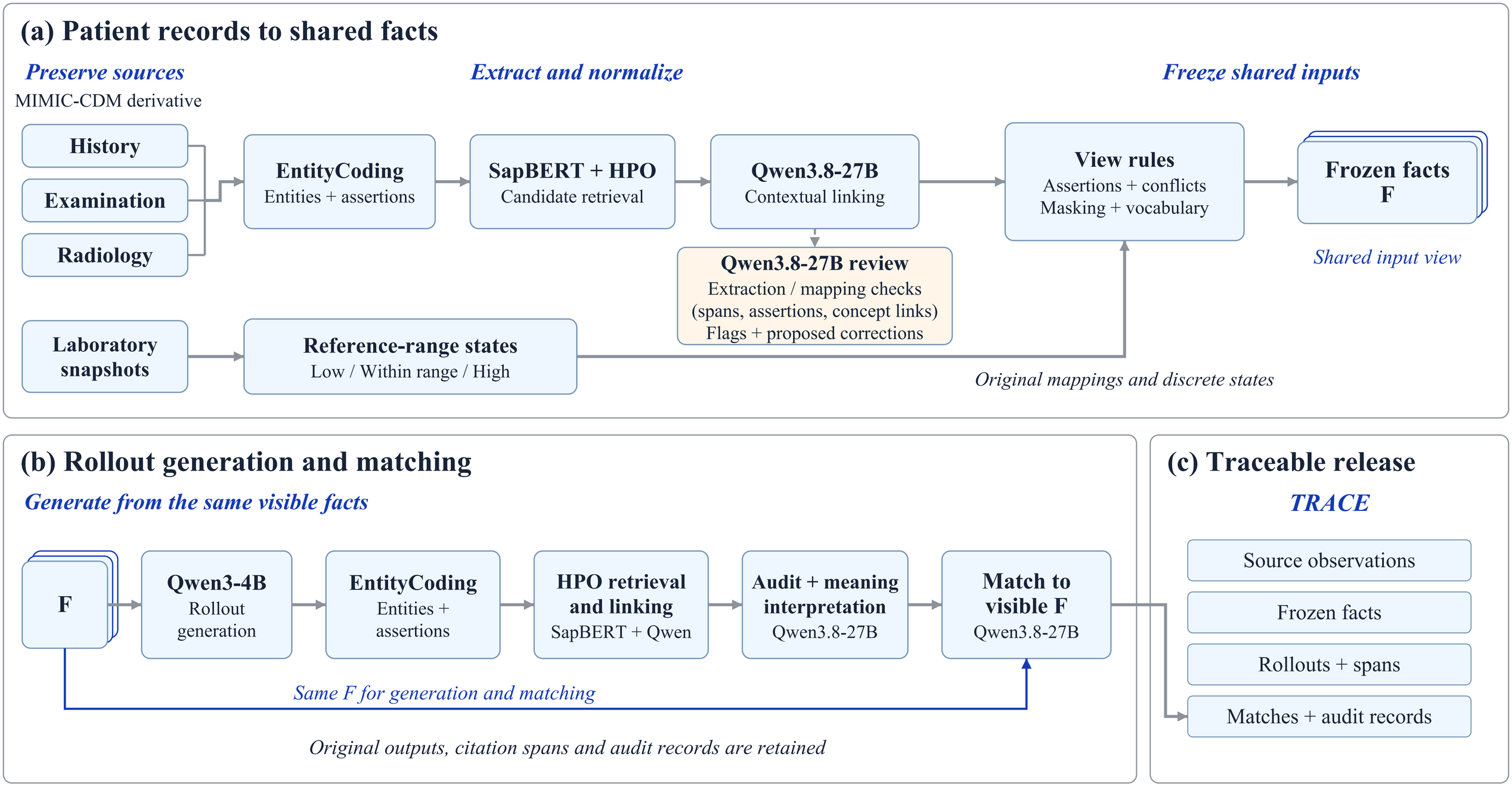

(a) Patient records to shared facts
Preserve sources
Extract and normalize
Freeze shared inputs
MIMIC-CDM derivative
History
Examination
Radiology
EntityCoding
Entities + assertions
SapBERT + HPO
Candidate retrieval
Qwen3.8-27B
Contextual linking
Qwen3.8-27B review
Extraction / mapping checks
(spans, assertions, concept links)
Flags + proposed corrections
View rules
Assertions + conflicts
Masking + vocabulary
Frozen facts
F
Shared input view
Laboratory snapshots
Reference-range states
Low / Within range / High
Original mappings and discrete states
(b) Rollout generation and matching
Generate from the same visible facts
F
Qwen3-4B
Rollout generation
EntityCoding
Entities + assertions
HPO retrieval and linking
SapBERT + Qwen
Audit + meaning interpretation
Qwen3.8-27B
Match to visible F
Qwen3.8-27B
Same F for generation and matching
Original outputs, citation spans and audit records are retained
(c) Traceable release
TRACE
Source observations
Frozen facts
Rollouts + spans
Matches + audit records